

MTH 166

Lecture-28

**Level Surfaces and Parametric
Equation of a Straight Line**

Unit 5: Vector Calculus-I

(Book: Advanced Engineering Mathematics by Jain and Iyengar, Chapter-15)

Topic:

Level Surfaces, Parametric equation of a Straight Line

Learning Outcomes:

1. To find Level surfaces of a given scalar function
2. To find parametric equation of a straight line

Level Surfaces:

Let $f(x, y, z)$ be a given scalar surface.

Level surfaces corresponding to this $f(x, y, z)$ are given by:

$$f(x, y, z) = c \quad (1)$$

In fact, this equation (1) gives family of surfaces that never intersect with each other

For different values of constant c , we get different members of this family of level surfaces.

Find the Level surfaces of the scalar fields defined by following functions:

Problem 1. $f = x + y + z$

Solution. The given scalar function is:

$$f = x + y + z \quad (1)$$

Level surfaces are given by: $f = c$

$$\Rightarrow x + y + z = c$$

Which is a family of Parallel planes.



Find the Level surfaces of the scalar fields defined by following functions:

Problem 2. $f = x^2 + y^2 + z^2$

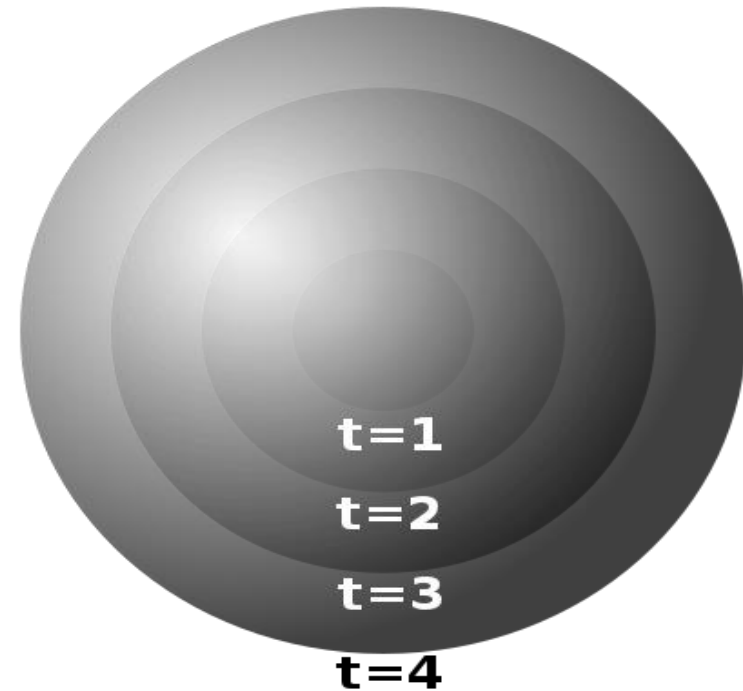
Solution. The given scalar function is:

$$f = x^2 + y^2 + z^2 \quad (1)$$

Level surfaces are given by: $f = t$

$$\Rightarrow x^2 + y^2 + z^2 = t$$

Which is a family of concentric Spheres.



Find the Level surfaces of the scalar fields defined by following functions:

Problem 3. $f = x^2 + y^2 - z$

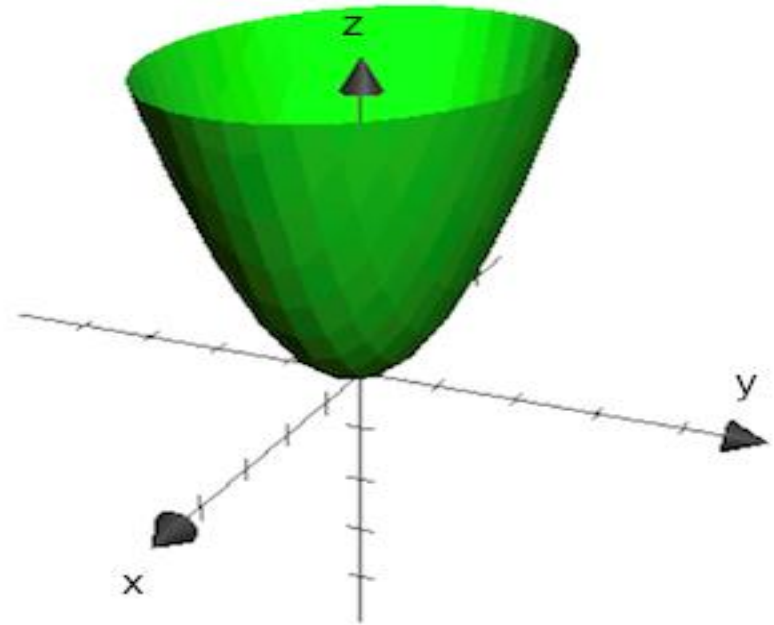
Solution. The given scalar function is:

$$f = x^2 + y^2 - z \quad (1)$$

Level surfaces are given by: $f = c$

$$\Rightarrow x^2 + y^2 - z = c$$

Which is a family of Paraboloids.



Find the Level surfaces of the scalar fields defined by following functions:

Problem 4. $f = x^2 + 9y^2 + 16z^2$

Solution. The given scalar function is:

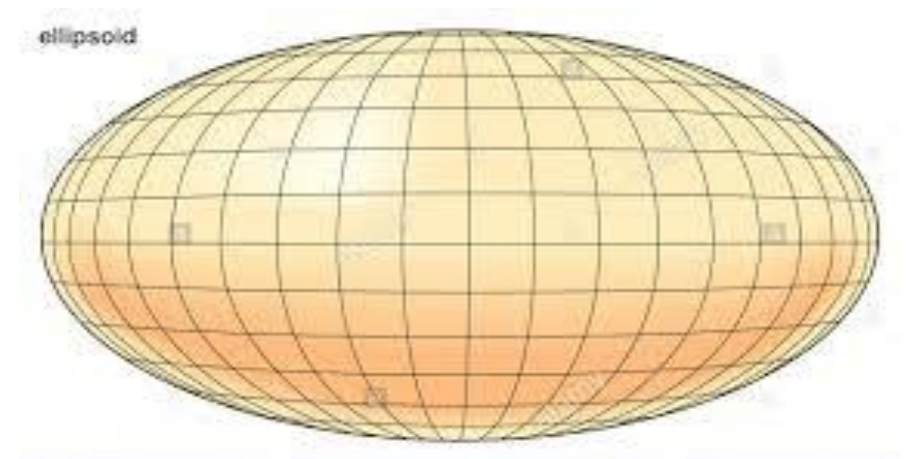
$$f = x^2 + 9y^2 + 16z^2$$

Level surfaces are given by: $f = c$

$$\Rightarrow x^2 + 9y^2 + 16z^2 = c$$

Which is a family of Ellipsoids.

(1)



Find the Level surfaces of the scalar fields defined by following functions:

Problem 5. $f = z - \sqrt{x^2 + y^2}$

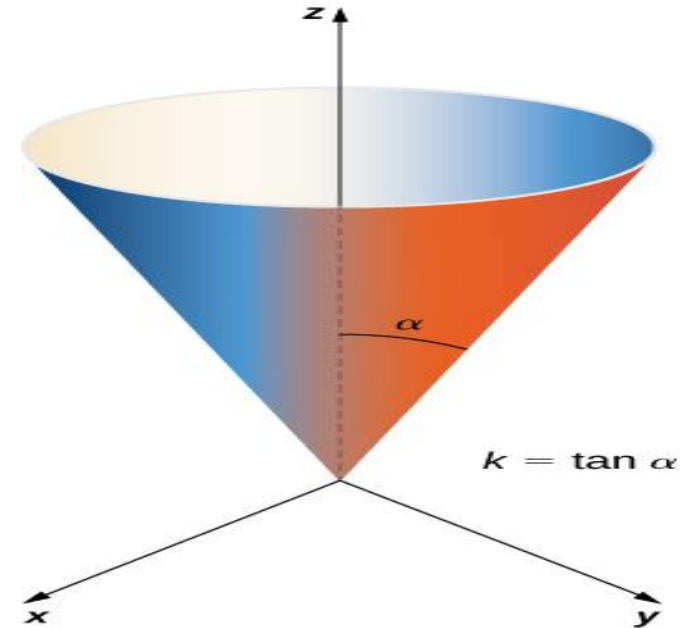
Solution. The given scalar function is:

$$f = z - \sqrt{x^2 + y^2} \quad (1)$$

Level surfaces are given by: $f = c$

$$\Rightarrow z - \sqrt{x^2 + y^2} = c$$

Which is a family of Cones.



Polling Quiz

The level surfaces of the scalar field defined by the function $f=x+y+z$ are:

- (a) Parallel Planes
- (b) Spheres
- (c) Paraboloids
- (d) Ellipsoids

Parametric Equation of a Straight line passing through a point and having a given direction:

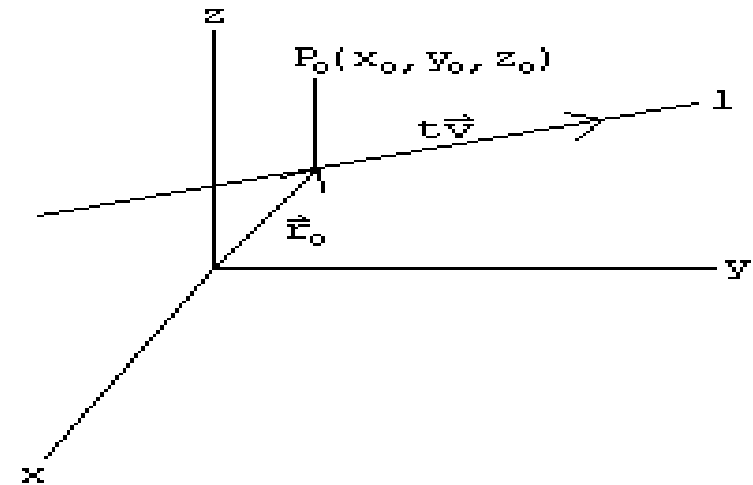
Let given point is $P_0(x_0, y_0, z_0)$ and the given direction be $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$

Then Parametric Equation of a Straight line passing through point P_0 and having given direction $\vec{v}$ is given by:

$$\vec{r}(t) = \vec{P}_0 + t \vec{v} \quad \text{where } t \text{ is parameter}$$

$$\Rightarrow \vec{r}(t) = (x_0\hat{i} + y_0\hat{j} + z_0\hat{k}) + t(v_1\hat{i} + v_2\hat{j} + v_3\hat{k})$$

$$\Rightarrow \vec{r}(t) = (x_0 + tv_1)\hat{i} + (y_0 + tv_2)\hat{j} + (z_0 + tv_3)\hat{k}$$



Find the Parametric Equation of a Straight line passing through point P_0 and having direction $\vec{b}$:

Problem 1. $P_0(1,2,3)$, $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$

Solution. Parametric Equation of a Straight line passing through point P_0 and having given direction $\vec{b}$ is given by:

$$\vec{r}(t) = \vec{P_0} + t \vec{b} \quad \text{where } t \text{ is parameter}$$

$$\Rightarrow \vec{r}(t) = (\hat{i} + 2\hat{j} + 3\hat{k}) + t(\hat{i} + 2\hat{j} + 3\hat{k})$$

$$\Rightarrow \vec{r}(t) = (1 + t)\hat{i} + (2 + 2t)\hat{j} + (3 + 3t)\hat{k} \quad \textbf{Answer.}$$

Find the Parametric Equation of a Straight line passing through point P_0 and having direction $\vec{b}$:

Problem 2. $P_0(1, -1, 1)$, $\vec{b} = \hat{i} - \hat{j}$

Solution. Parametric Equation of a Straight line passing through point P_0 and having given direction $\vec{b}$ is given by:

$$\vec{r}(t) = \vec{P_0} + t \vec{b} \quad \text{where } t \text{ is parameter}$$

$$\Rightarrow \vec{r}(t) = (\hat{i} - \hat{j} + \hat{k}) + t(\hat{i} - \hat{j})$$

$$\Rightarrow \vec{r}(t) = (1 + t)\hat{i} - (1 + t)\hat{j} + \hat{k} \quad \textbf{Answer.}$$

Polling Quiz

The parametric representation of the straight line through the point $P(1,2,3)$ and having direction $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$ is:

(a) $(\vec{r}(t)) = (1+t)\hat{i} + 2(1+t)\hat{j} + (3+2t)\hat{k}$

(b) $(\vec{r}(t)) = (1+t)\hat{i} - (1+t)\hat{j} + \hat{k}$

(c) $(\vec{r}(t)) = (1+t)\hat{i} - 2(1+t)\hat{j} + (3-2t)\hat{k}$

(d) $(\vec{r}(t)) = (1+t)\hat{i} + (1+t)\hat{j} - \hat{k}$

Find the parametric representation of the following straight lines/curves. Use the indicated representation wherever given:

Problem 1. $x = y, y = z$

Solution. Let $z = t$ where t is a parameter

$$\Rightarrow y = t \quad (\because y = z)$$

$$\text{also } x = t \quad (\because x = y)$$

So, Parametric representation is given by:

$$\overrightarrow{r(t)} = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

$$\Rightarrow \overrightarrow{r(t)} = t\hat{i} + t\hat{j} + t\hat{k} \quad \textbf{Answer.}$$

Problem 2. $x + y + z = 3, y - z = 0$.

Solution. Let $z = t$ where t is a parameter

$$\Rightarrow y = t \quad (\because y = z)$$

also $x + y + z = 3$

$$\Rightarrow x = 3 - y - z = 3 - t - t$$

$$\Rightarrow x = 3 - 2t$$

So, Parametric representation is given by:

$$\overrightarrow{r(t)} = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

$$\Rightarrow \overrightarrow{r(t)} = (3 - 2t)\hat{i} + t\hat{j} + t\hat{k} \quad \textbf{Answer.}$$

Problem 3. $y^2 + z^2 = 9, x = 9 - y^2, y = 3 \sin t$.

Solution. Since $y = 3 \sin t$

$$\Rightarrow x = 9 - (3 \sin t)^2$$

$$\Rightarrow x = 9(1 - \sin^2 t) = 9\cos^2 t$$

$$\text{also } z^2 = 9 - y^2 = 9 - (3 \sin t)^2 = 9(1 - \sin^2 t) = 9\cos^2 t$$

$$\Rightarrow z = \pm 3 \cos t$$

So, Parametric representation is given by:

$$\overrightarrow{r(t)} = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

$$\Rightarrow \overrightarrow{r(t)} = (9\cos^2 t)\hat{i} + (3 \sin t)\hat{j} \pm (3 \cos t)\hat{k} \quad \textbf{Answer.}$$

Problem 4. $y^2 = x^2 + z^2, y = 2, x = 2 \sin t$.

Solution. Since $x = 2 \sin t$ and $y = 2$

$$z^2 = y^2 - x^2$$

$$\Rightarrow z^2 = 4 - (2 \sin t)^2 = 4(1 - \sin^2 t) = 4\cos^2 t$$

$$\Rightarrow z = \pm 2 \cos t$$

So, Parametric representation is given by:

$$\overrightarrow{r(t)} = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

$$\Rightarrow \overrightarrow{r(t)} = (2 \sin t)\hat{i} + 2\hat{j} \pm (2 \cos t)\hat{k} \quad \textbf{Answer.}$$

